

Noether Sheet 7

Jonathan Kasongo

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These were my solutions to the Noether Mentoring Scheme Sheet 7. Problem statements are abridged.

Problem 1

Show that for real numbers x, y and z we have

$$x(x - y) + y(y - z) + z(z - x) \geq 0.$$

Find all cases where equality occurs.

Solution

We want to show $x^2 + y^2 + z^2 \geq xy + yz + zx$.

Theorem — Cauchy-Schwarz Inequality: For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$\mathbf{u} \cdot \mathbf{v} \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

with equality if and only if the two vectors are parallel.

Proof: It follows very simply from the formula for the dot product

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between the two vectors. Now since $\cos \theta \leq 1$ we have $\mathbf{u} \cdot \mathbf{v} \leq \|\mathbf{u}\| \|\mathbf{v}\|$. The equality case happens only when $\cos \theta = 1$ which means $\theta = 0^\circ$, that is the two vectors are parallel. $\square$

Now let us apply Cauchy-Schwarz with the vectors $\mathbf{u} = \langle x, y, z \rangle$ and $\mathbf{v} = \langle y, z, x \rangle$. We have

$$xy + yz + zx \leq \sqrt{x^2 + y^2 + z^2} \sqrt{y^2 + z^2 + x^2} = x^2 + y^2 + z^2$$

which is the desired inequality. The equality case occurs if $\mathbf{u}$ and $\mathbf{v}$ are parallel which happens only if

$$\frac{x}{y} = \frac{y}{z} = \frac{z}{x}, \quad (x, y, z \text{ all non-zero})$$

$$(x, y, z) = (0, 0, 0), \quad (\text{otherwise}).$$

with the second case being due to the following argument. Suppose WLOG that $z = 0$. Since the third component of $\mathbf{u}$ and $\mathbf{v}$ have to be in constant ratio we require $x = 0$. Since the second components have to be in constant ratio this forces $y = 0$, and so all three variables must be 0. $\blacksquare$

Remark: This is just a special case of the Re-arrangement inequality.

Addendums, (after having read solutions): I believe that my solution is completely correct, because actually, for x, y, z all non-zero the following is true

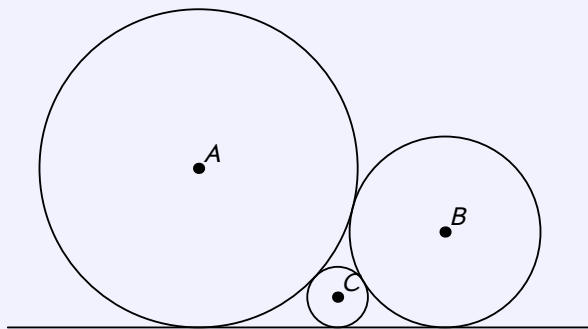
$$\frac{x}{y} = \frac{y}{z} = \frac{z}{x} \iff x = y = z$$

like how it was presented in the official solutions.

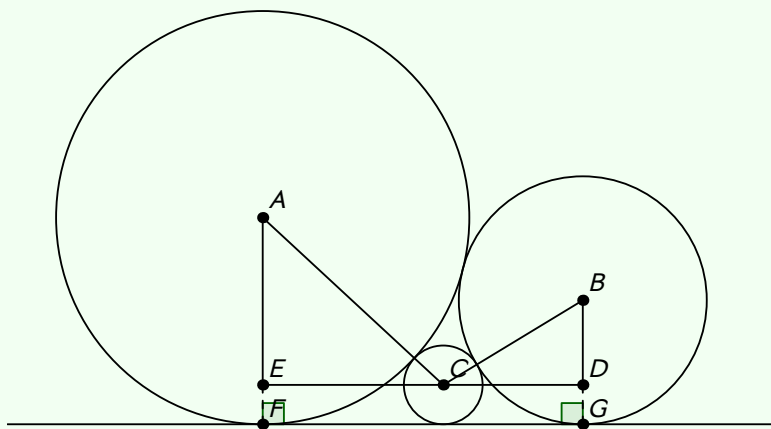
Problem 2

Three circles touch each other and a line as shown. Prove that the radii a, b, c where c is the smallest, satisfy the equation

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$$

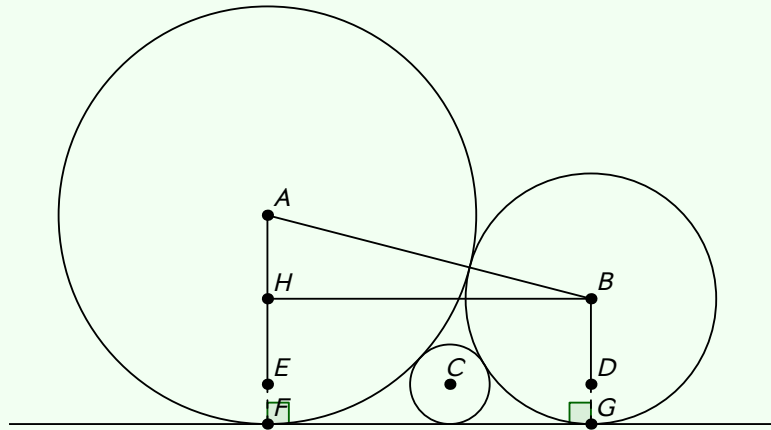
**Solution**

Let the circles have centres A, B, C with radii a, b, c respectively. The idea is to draw in the following lines,



First look at $\triangle CBD$. Side $BC = b + c$ since the two circles are tangent to each other. Side $BD = BG - DG = b - c$. Thus by Pythagoras $CD = \sqrt{(b + c)^2 - (b - c)^2} = 2\sqrt{bc}$.

Similarly look at $\triangle CEA$. Side $AC = a + c$ since the two circles are tangent. Side $AE = AF - EF = a - c$. Thus by Pythagoras $CE = \sqrt{(a + c)^2 - (a - c)^2} = 2\sqrt{ac}$. So putting these two together, $FG = ED = 2\sqrt{bc} + 2\sqrt{ac}$. Now consider the following



Look at $\triangle AHB$. Side $AB = a + b$ since the two circles are tangent. Side $AH = AF - HF = a - b$. So by Pythagoras, $HB = \sqrt{(a + b)^2 - (a - b)^2} = 2\sqrt{ab}$. That is also equal to FG . So we deduce

$$2\sqrt{ab} = 2\sqrt{bc} + 2\sqrt{ac}$$

and dividing by $2\sqrt{abc}$ yields the desired result,

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}. \blacksquare$$

Addendums (after having read solutions): A classical result that comes up fairly often usually in multiple choice type math contests. I have seen this idea before.

Problem 3

Find all quadruples (a, b, c, d) of positive integers such that $a \leq b \leq c \leq d$ and

$$a + b + c + d = ab + cd.$$

Solution

Due to this ordering condition we notice that

$$a + b + c + d \leq d + d + d + d = 4d.$$

But since this sum is the same as $ab + cd$ we note that $ab + cd \leq 4d$. Now if $c \geq 4$ the quantity $ab + cd$ will be too large, so $c < 3$.

If $c = 3$, we have

$$a + b + 3 + d = 3d + ab$$

$$a + b - ab + 3 = 2d$$

If $b = 3$ then

$$-2a + 6 = 2d$$

$$-a + 3 = d$$

but this implies that $d < 3 = c$ which is a contradiction.

If $b = 2$ then

$$-a + 5 = 2d$$

$$\frac{5 - a}{2} = d$$

but this implies that $d < 2.5 < c$, a contradiction.

If $b = 1$, then this forces $a = 1$ due to the ordering condition. We have

$$1 + 2 + 3 + d = 2 + 3d$$

$$4 = 2d$$

so $d = 2$. But $d < 3 = c$ is a contradiction.

If $c = 2$, we have

$$a + b + 2 + d = ab + 2d$$

$$a + b - ab + 2 = d$$

If $b = 2$ then

$$-a + 4 = d$$

If $a = 1$ then $d = 3$ which yields a solution $(a, b, c, d) = (1, 2, 2, 3)$.

If $a = 2$ then $d = 2$ which yields a solution $(a, b, c, d) = (2, 2, 2, 2)$.

If $a > 2$ then $d < 2 = c$ which is a contradiction.

If $c = 1$, we have $a = b = c = 1$ due to the ordering condition. This gives $3 + d = 1 + d$ which clearly has no solutions.

Thus the only solutions are $(a, b, c, d) = (1, 2, 2, 3), (2, 2, 2, 2)$. ■

Addendums (after having read solutions): On this problem I simply just forgot the case if $c = 2$ and $b = 1$ and this yields the final solution $(1, 1, 2, 3)$.

Problem 4

- (a) How many different values is it possible to make using three different British coins?
- (b) In how many ways can three numbers be chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ such that no two of them are consecutive?
- (c) You should notice that these two numbers are the same. Why is this?

Preliminaries

Preliminaries: Let $f : A \rightarrow B$ be a function. The set of inputs A is the domain of f . The set of all outputs B is the co-domain of f or also known as the image of f . I may use both terminology.

An injective function $f : A \rightarrow B$ is a function with the following property.

$$f(x_1) = f(x_2) \iff x_1 = x_2$$

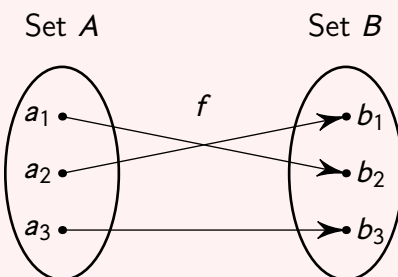
Roughly speaking that means that exactly one unique input can map to a one unique output. For example $f(x) = \ln(x)$ is injective whilst $f(x) = x^2$ isn't.

A surjective function is a function $f : A \rightarrow B$ is a function with the following property.

$$\text{For every } b \in B, \text{ there exists some } a \in A \text{ such that } f(a) = b$$

Roughly speaking that means that the function hits every element in the co-domain. For example $f(x) = 2x$ is surjective whilst $f(x) = e^x$ isn't (if we assume the codomain is the entirety of $\mathbb{R}$).

A bijective function is a function $f : A \rightarrow B$ which is both injective and surjective. Bijective functions are also known as "one to one" functions because every unique element in the domain A is paired up with it's own unique element in the image B . I will skip out the formal definition here (because im lazy) but the following diagram gives us a good intuition for the idea of a bijective function.



Solution

(a) The answer is just $\binom{8}{3} = 56$ since each triple of coins produces it's own unique sum. We prove this statement by showing that $f(x, y, z)$ is an injection, where f is defined below.

The eight British coins have values: 1p, 2p, 5p, 10p, 20p, 50p, £1, £2. It is nicer to think about these values all in terms of pence. Define the sequence

$$(a_n)_{n=1}^8 = (1, 2, 5, 10, 20, 50, 100, 200).$$

For any triple of integers (x, y, z) with $1 \leq x < y < z \leq 8$, define $f(x, y, z) = a_x + a_y + a_z$.

Claim 1: Let $2 \leq n \leq 8$ be an integer. Then $a_1 + a_2 + \dots + a_{n-1} < a_n$.

Proof: Check each case manually. $\square$

Claim 2: The function f is injective.

Proof: Suppose FTSOC that f is not injective. Then there exists some triple (x, y, z) with the following property. There exists some different triple (u, v, w) such that $f(x, y, z) = f(u, v, w)$. Now aiming for a contradiction we are going to show that (x, y, z) and (u, v, w) are in fact the same triple.

First we show that $w = z$, by showing that $w > z$ and $w < z$ both give contradictions. Note that $3 \leq w, z \leq 8$ due to ordering conditions.

- Suppose that $w > z$. Because claim 1 tells us that any sum S of distinct elements that come before a_w in the sequence, satisfies $S < a_w$, we have $f(x, y, z) < a_w < f(u, v, w)$ a contradiction.
- Suppose that $w < z$. Similarly because claim 1 tells us that any sum S of distinct elements that come before a_z in the sequence, satisfies $S < a_z$, we have $f(u, v, w) < a_z < f(x, y, z)$ a contradiction.

Thus the only possibility is that $w = z$. Now we are going to show that $v = y$ using the same strategy. Because $w = z$ our initial assumption reduces to $a_u + a_v = a_x + a_y$. Note that $2 \leq v, y \leq 7$ due to ordering conditions.

- Suppose that $v > y$. Because claim 1 tells us that any sum S of distinct elements that come before a_v in the sequence satisfies $S < a_v$, we have $a_x + a_y < a_v < a_u + a_v$ a contradiction.
- Suppose that $v < y$. Because claim 1 tells us that any sum S of distinct elements that come before a_y in the sequence, satisfies $S < a_y$, we have $a_u + a_v < a_y < a_x + a_y$ a contradiction.

Thus the only possibility is that $v = y$. So now our initial assumption reduces to $a_u = a_x$. Since there are no repeated values in (a_n) it must be the case that $u = x$ and so in fact (x, y, z) and (u, v, w) are the same exact triple, contradiction. $\square$

Now due to claim 2, each selection of three distinct British coins produces a unique sum, so the number of different values one can make is just the number of unordered triples of three coins without repetition and that works out to be

$$\binom{8}{3} = 56. \quad \square$$

(b) We will use complementary counting. Call an unordered triple (c, b, a) of integers $1 \leq c < b < a \leq 10$ nice.

We want to count the number N of nice triples that have at least one pair of consecutive numbers. For each pair $(n, n + 1)$, $(1 \leq n \leq 9)$ there are 8 other numbers that we can put in the triple giving $9(8)$ such nice triples. However we have double counted all the nice triples of the form $(n, n + 1, n + 2)$, $(1 \leq n \leq 8)$ since they have two pairs of consecutive numbers. So in fact $N = 9(8) - 8 = 64$.

Thus the desired count works out as

$$\binom{10}{3} - 64 = 56. \quad \square$$

(c) Call a nice triple *super nice* if the triple has no pair of consecutive numbers. Call a nice triple (x, y, z) *almost nice* if it satisfies $(2 \leq x < y < z \leq 9)$. There are exactly $\binom{8}{3}$ almost nice triples.

I am going to show that there is a bijection (a "perfect" one to one correspondence) between super nice triples and almost nice triples, and this will force our conclusion "The number of super nice triples is the same as the number of almost nice triples. This is why there are $\binom{8}{3}$ super nice triples."

Let f be the bijection. Precisely we have the following

$$f : \{\text{Super nice triples}\} \rightarrow \{\text{Almost nice triples}\}.$$

Now we will describe the map f . Consider some arbitrary super nice triple (p, q, r) . Note that $3 \leq q \leq 8$ due to the ordering restrictions on super nice triples. We will map it to the almost nice triple (x, q, z) with the following rules.

- If $p = 1$ then $x = q - 1$. Otherwise $x = p$.
- If $q = 10$ then $z = q + 1$. Otherwise $z = r$.

With the inequality $3 \leq q \leq 8$ it can be checked that the output triple (x, q, z) is indeed almost nice. Since we have found a bijection between super nice triples and almost nice triples, the number of triples of both forms is the exact same and this is why there are $\binom{8}{3} = 56$ super nice triples. ■

Addendums (after having read solutions): I believe that my solutions to (a) and (b) are correct and justified. The hard part is part (c), it's quite difficult I feel to communicate one's solution clearly for this one.

I hope that you can at least understand the gist of my solution, if I'm being perfectly honest I didn't really understand the official solution here I feel that it lacks a lot of rigour.

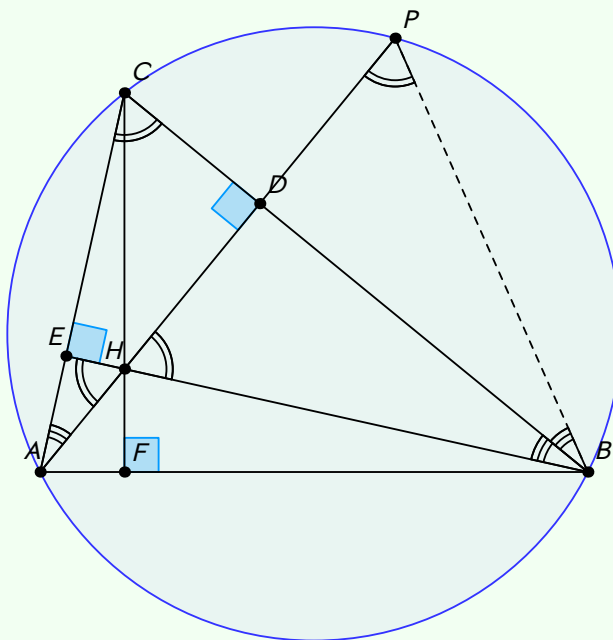
Problem 5

An acute-angled triangle ABC is inscribed inside a circle. The altitudes of the triangle AD, BE, CF meet at H , the orthocentre of the triangle.

- (a) If line AD is extended to meet the circle at P , prove that $HD = DP$.
 (b) Prove that H is the incentre of triangle DEF .

Solution

(a) Here's a diagram.

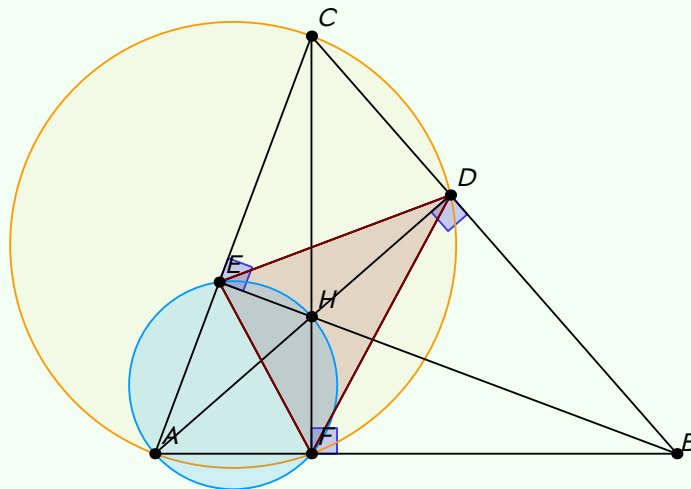


First let's study $\triangle CDA$. Call $\angle DCA = \theta$. Then $\angle CAD = 90 - \theta$. Now let's look at $\triangle EHA$. We have $\angle EHA = \theta$ since $\angle HEA = 90$. Now due to vertically opposite angles $\angle DHB = \angle EHA = \theta$.

However due to the "Angles subtending the same segment are equal" theorem we also have $\angle APB = \angle ACB = \theta$ or equivalently $\angle ACD = \theta = \angle DPB$. Now notice that $\angle HDB = \angle PDB = 90$ due to vertically opposite angles + angles on a straight line sum to 180. Moreover observe that $\triangle DPB$ and $\triangle DBH$ share two pairs of equal angles, and since angles in a triangle sum to 180, their third angles are equal also. Thus the two triangles are similar by AAA criterion.

But since the two triangles share a side DB , they are actually congruent to each other by ASA criterion. This triangle congruence implies the desired result $DH = DP$. $\square$

(b) Here's a diagram. I will show that CF bisects $\angle EFD$, and then one can use a similar process to show the other bisections.



Claim 1: AFDC is cyclic.

Proof: Look at $\triangle BDA$. We have $\angle BDA = 180 - (90 + \angle B) = 90 - \angle B$. Look at $\triangle FCB$. We have $\angle FCB = 180 - (90 + \angle B) = 90 - \angle B$. Now since $\angle BDA = \angle FCB$ both subtend the segment DF , by the converse of the "Angles subtending the same segment are equal" theorem we know that A, F, D, C are concyclic. $\square$

Claim 2: AFHE is cyclic.

Proof: We have opposite angles summing to 180 since $\angle AEH = \angle AFH = 90$. These two summing to 180 forces the other pair of opposite angles to sum to 180 also, because angles in a quadrilateral sum to 360. Thus $AFHE$ is cyclic. $\square$

By angles subtending the same segment are equal on $AFDC$ we have $\angle CFD = \angle CAD$. By the same theorem on $AFHE$ we have $\angle EAH = \angle EFH$, or equivalently $\angle CAD = \angle EFH$. Now since $\angle CFD = \angle EFH$ the line CF has in fact bisected $\angle EFD$. $\blacksquare$

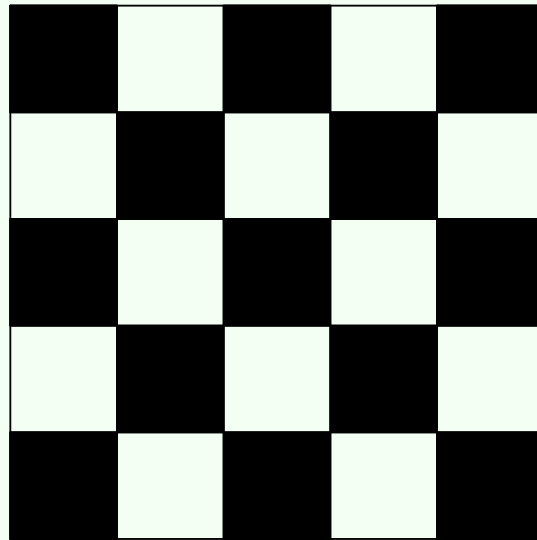
Addendum (after having read solutions): I think part (a) is fine here. For part (b) I only showed that CF bisects $\angle EFD$ out of laziness, I kinda just hoped that the same method will work to show the two other bisections (I don't see any reason why it should work out that way).

Problem 6

- (a) Suppose twelve 2×1 dominoes are placed on a 5×5 chessboard such that each covers exactly two squares. Which squares can possibly be left uncovered?
- (b) Suppose instead that eight 3×1 triominoes are used, each covering exactly three squares. Which squares can be left uncovered now?
- (c) What if six 4×1 tetrominoes are used, with each covering four squares. Which squares can be left uncovered in that case?

Solution

- (a) There are two ways that a 5×5 chessboard could be coloured, with 12 black squares and 13 white squares or with 13 black squares and 12 white squares. WLOG let us assume the latter case,



Each of the twelve 2×1 dominoes covers exactly one white square and one black square. All together they will cover 12 white squares and 12 black squares. Thus the one square left uncovered must be black. Now we are going to show that any black square could be left uncovered. We show that there is an algorithm that allows us to always cover the board.

Algorithm 1.

We are going to tile a 5×5 chessboard with one black square removed. We are going to dissect the shape into twelve 2×1 pieces.

1. Consider the topmost row of the board, now take the leftmost square on this row, call the square s .
2. If there is a square just below s then, remove both s and this square from the board. Otherwise take the square to the right of s and remove both s and this square from the board.
3. If the board still has squares left go back to step 1, otherwise terminate the algorithm here.

Now we are going to prove algorithm 1 always works to tile the board.

Proof of correctness: We need to show that the square below or to the right of s always exists in step 2. Consider the square s that we refer to in step 2. The following is a key lemma in our argument, call it (*). Observe that since we always take s to be the topmost then the leftmost square available, the previously discarded squares were either above s or on the left of s .

Case 1: s is black. If s is black, and if s doesn't lie on the bottom row, the square below it b , is white. The square b exists because it wasn't previously discarded due to (*), and it isn't the one square that was removed from the chessboard, since we removed a black square from the board.

Now if s does lie on the bottom row then s has a square to the right of it unless s happens to be the square in the very bottom right of the chessboard. But since s is the topmost then the leftmost square available, this must mean that s is the only square left on the entire chessboard at the moment. However since there were initially $24 \equiv 0 \pmod{2}$ squares, and each time we are removing $2 \equiv 0 \pmod{2}$ squares from the shape, the number of squares currently part of the shape N , satisfies the invariant $N \equiv 0 \pmod{2}$. Thus s cannot be the only square left, a contradiction.

Case 2: s is white. If s is white, and not on the bottom row, then there is a black square below it b , only if b was not the square we chose to remove from the board. Suppose that b was in fact the square we chose to remove. If s doesn't lie on the right most column then we can just pick the black square to the right of s . Suppose that s actually lies on the right most column. Due to our diagram configuration s doesn't lie on the topmost row, so there was a black b' square above s previously. Since b' lies on the rightmost column it must have been paired up with the square below it, s , but this means that s doesn't exist a contradiction.

Now suppose s does lie on the bottom row. Due to our diagram configuration there is always a black square b to the right of s . The only issue occurs when this black square b happens to be the one removed from the chessboard. If this is the case then there was a black square above s call it b' . At that point in time the square below b' was available to take and since our algorithm first prioritises taking the square below b' we would have removed b' and s previously. But this means that s doesn't exist, a contradiction. $\square$

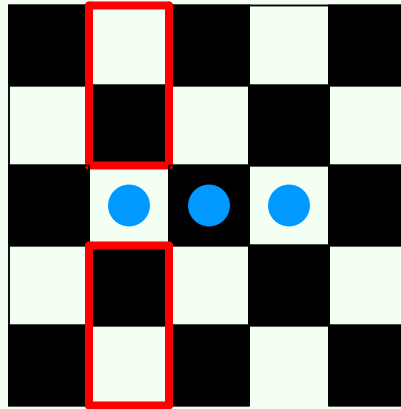
(b) After experimenting we find that it seems to be that only the very middle square can be left uncovered. Let's prove our guess.

Suppose FTSOC that the middle square has been covered, by (WLOG) a horizontal 3×1 triomino. The following claim will be useful to us.

Claim 1: For any row, at most one horizontal 3×1 triomino can be placed there. Similarly for any column, at most one vertical 1×3 triomino can be placed there.

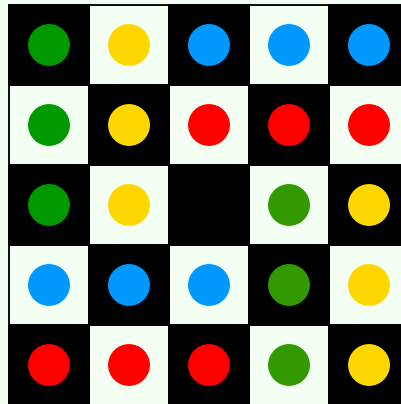
Proof: We will show the first statement to be true, this also gives the second statement to be true since we can rotate the board by 90° . In a row there are 5 squares. Each 3×1 triomino uses exactly 3 squares. If we could place $t \geq 2$ triominos on a row then we require $3t \leq 5$ but since $t \geq 2$ this inequality is false. Thus $t \leq 1$ as desired. $\square$

The horizontal triomino that we have placed to cover the middle square also splits precisely 3 columns into 1×2 rectangles that are above and below of our triomino. The diagram below illustrates the idea.



This means for those three columns, there is no longer any way to place a vertical 1×3 triomino. Now using claim 1, on the entire board we can have at most 5 horizontal triominos, and at most $5 - 3 = 2$ vertical triominos. In total we can place at most 7 triominos but we wanted to place 8 triominos, a contradiction.

Now we show a way to place the 8 triominoes with just the very middle uncovered. $\square$



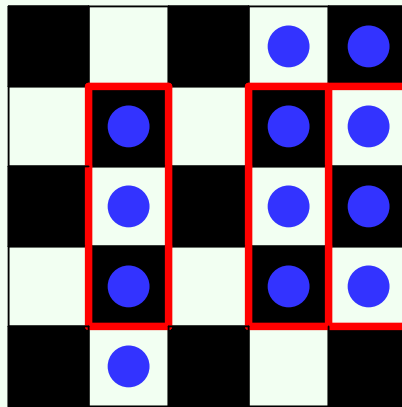
(c) Again after experimenting we find that only the corners of the board can be left uncovered in this case. Let's prove our guess.

Similar to claim 1 (b), we can place at most 5 horizontal (or vertical) 4×1 tetrominoes on the board, because each row (or column) only has enough squares to accommodate for at most one 4×1 tetromino. Suppose we use v vertical tetrominoes and h horizontal tetrominoes. At least one of v, h has to be ≥ 3 because otherwise

$$6 = v + h < 3 + 3 = 6$$

is a contradiction. Suppose WLOG that $v \geq 3$.

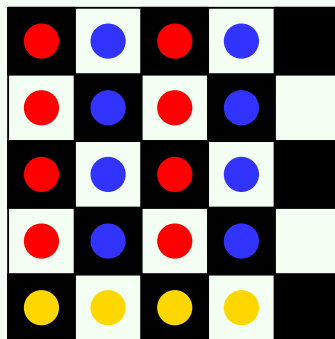
Each of those vertical pieces always covers the middle three squares of that column. A picture helps to illustrate.



So now we can no longer place horizontal tetrominos along those middle three rows because there are only two squares left uncovered whilst we need at least 4 to place our tetrominos. Thus we must place horizontal tetrominos along the top or the bottom of the board.

Actually we can place at most one of these horizontal tetrominos on the board since if we place two then we will be unable to place any vertical tetrominoes on the central three columns, by a similar argument to the one above. Thus in order to place 6 tetrominos we must use 5 vertical tetrominoes and 1 horizontal tetromino.

There must be a vertical tetromino in every column, since we have a 5×5 board. Once we place the horizontal tetromino along the top or the bottom, in each of the columns that the horizontal tetromino passes through, there is now only one way to place the vertical tetromino for that column. The only degree of freedom we have is where we place the last vertical tetromino.



in any case wherever we place the last vertical tetromino the uncovered square will be in the corner. Finally it is easy to demonstrate by construction that every corner can in fact be left uncovered (I omit diagrams here because I'm tired of dealing with Geogebra by now. :p). ■

Remarks: Interestingly enough, for me at least, part (c) was the easiest, then part (b), then part (a) was the hardest, whilst usually it's the reverse way round in terms of difficulties.

Addendums (after having read solutions): This is a nice game-theory type problem. Each of my three solutions look really different to the official solutions so let's discuss them one by one.

For part (a), I remembered the last problems we did on chessboards like this and that it should be possible to just remove any black square from the board and then tile the subsequent board with 2×1 dominoes. Playing around with the problem I found that there are many ways to do these tilings so I just guessed that a greedy solution might work. Then finally devising the algorithm and proving its correctness feels like a standard codeforces problem. The case bashing in the official solutions is probably a much clearer solution.

For parts (b) and (c) my two solutions are pretty similar, somehow some configuration must be forced then we case bash to show the final result. The colouring certain tiles solution from the official sols here are particularly elegant.

Problem 7

Let us call a positive integer n "mischievous" if both n and $n + 1$ are divisible by perfect cubes greater than 1. Show that at least 1% of positive integers are mischievous.

You might wonder how you can talk about 1% of an infinite number of numbers. What is meant is that once you take sufficiently many numbers, at least 1% are mischievous. Formally, you want to show that there is some N such that any set of N consecutive numbers, then 1% of them are mischievous.

Initial thoughts and motivations

Motivations. Let p, q, n be positive integers. Employing some wishful thinking it would be nice if I could show that there is a family of solutions to

$$p^3 - nq^3 = 1$$

because then I could set $n + 1 = nq^3 + 1$, $n = nq^3$ and this way both of them are divisible by perfect cubes. Looking at this equation it looks similar to the usual Pell equation form but instead of squares we have cubes.

Next I naturally thought about the identity

$$n^3 + 1 = (n + 1)(n^2 - n + 1)$$

since it looks similar to what I wanted above. Actually if I set $n + 1 = t^3$ for some $t \in \mathbb{N}$ then this would be exactly what I wanted and I believe it actually works. Let's have a go.

Let $n \in \mathbb{N}$. Consider the following fact

$$n^3 + 1 = (n + 1)(n^2 - n + 1)$$

Let $k \in \mathbb{N}$. Set $n = k^3 - 1$. Now we have

$$(k^3 - 1)^3 + 1 = k^3 [(k^3 - 1)^2 - (k^3 - 1) + 1].$$

Thus since $(k^3 - 1)^3 + 1$ is divisible by a perfect cube, the number $(k^3 - 1)^3$ is mischievous.

Unfortunately although this shows that there are infinitely many mischievous numbers, the gap between two consecutive numbers of this form shoots off to infinity for large enough k , so this approach won't work to show anything about the density of mischievous numbers. Circling back to the pell equation approach, we could consider the equation

$$x^6 - ny^6 = 1$$

but analysing the density of these solutions seems more complicated than I'd hope, so I think it's better to look for something else. If I can factor the number $kn^3 + 1$ into (cube)(something), then this would be desirable. I wondered what if for that (cube) I just try and force $kn^3 + 1$ to be divisible by some small cube say 8. If you go down this path you can show, for instance, that the numbers $7^3, 15^3, 23^3, \dots$ are all mischievous. These numbers are denser than the last family, this seems like progress. Now what if I replaced 8 with 27. Then you can show that the numbers $26^3, 53^3, 80^3, \dots$ are all mischievous. Let's try and generalise. I conjecture that all numbers of the form

$$(k^3 - 1 + tk)^3, \quad (t = 0, 1, 2, \dots)$$

are all mischievous. Still though I think these numbers aren't dense enough. An equivalent formulation of my idea is to study the modular equation

$$kn^3 + 1 \equiv 0 \pmod{p^3}$$

$$kn^3 \equiv -1 \pmod{p^3}$$

Let's restrict our analysis to say $p = 2$. It might be helpful to employ Euler's theorem here, so we compute $\varphi(8) = 8(1 - 1/2) = 4$. Then for $\gcd(n, 8) = 1$ we can write,

$$kn^3 \equiv -1 \pmod{8}$$

$$kn^4 \equiv -n \pmod{8}$$

$$k \equiv -n \pmod{8}$$

So actually pairs of the form $(k, n) = (-n, n) \pmod{8}$, $\gcd(n, 8) = 1$ should always give us mischievous numbers. Numbers of this class still aren't dense enough so maybe let's try to generalise. We note that $\varphi(p^3) = p^3(1 - 1/p) = p^2(p - 1)$.

$$kn^3 \equiv -1 \pmod{p^3}$$

$$kn^3 \cdot n^{p^3-p^2-3} \equiv -n^{p^3-p^2-3} \pmod{p^3}$$

$$k \equiv -n^{p^3-p^2-3} \pmod{p^3}$$

Thus pairs of the form

$$(k, n) = (-n^{p^3-p^2-3}, n) \pmod{p^3}$$

always produce mischievous numbers for $\gcd(p, n) = 1$. Modulo p^3 the number $kn^3 + 1$ then looks like

$$-n^{p^3-p^2} + 1 \equiv n^{p^3-p^2} - 1 \equiv 0$$

and we can make the statement that numbers of the form

$$n^{p^3-p^2} - 1, \quad (n = 1, 2, \dots, p-1)$$

are always mischievous. There is the corollary that for instance numbers of the form $n^4 - 1$ are also always mischievous, for $\gcd(n, 8) = 1$. Roughly we can say that the gap between two consecutive numbers of the form $n^4 - 1$ grows like $O(n^3)$. More generally the gap between two consecutive numbers of the form $n^{p^3-p^2} - 1$, grows like $O(n^{p^3-p^2-1})$. If I can improve this gap to even $O(\text{some large constant})$ this would be great. Actually I see a way to improve this gap to $O(n^2)$. Here is the motivation.

I seek to find some number $n = kp^3 - 1$ with the condition $v_q(kp^3 - 1) \geq 3$ for some prime q . If swap k with k^3 I could invoke LTE here. In particular if I let k, p be odd the following happens via LTE

$$v_2(k^3p^3 - 1) = v_2(kp - 1)$$

Now if I let $kp - 1 = 8x$ this guarantees $v_2(kp - 1) \geq 3$ and thus numbers of the form $(8x + 1)^3 - 1$ are mischievous. Actually re-reading over this that statement is quite obviously true since reducing mod 8 gives $1 - 1 = 0$. Moreover it is indeed the case that the gap between two consecutive numbers of this form grows like $O(x^2)$ which is an improvement! We can generalise this idea to get that numbers of the form $(p^3x + 1)^3 - 1$ are mischievous also, where p is a fixed prime.

From here it is natural to try and improve the gap to linear or constant. Another idea is to study the solutions to

$$27x + 8y = 1.$$

We can find solutions by say using Bézout's theorem. After playing around with the equation we notice the following pattern

$$\begin{aligned}
 27(3) - 8(10) &= 1 \\
 27(3) - 8(10) - 27(8) + 27(8) &= 1 \\
 27(11) - 8(37) &= 1 \\
 27(11) - 8(37) - 27(8) + 27(8) &= 1 \\
 27(19) - 8(64) &= 1 \\
 &\dots
 \end{aligned}$$

and so we find that the numbers $8(10), 8(37), 8(64), \dots$ are all mischievous, since adding 1 gives a number divisible by 27. Now this is very promising since these numbers are much more dense than the numbers we were considering before. If this conjecture is true then we will get that all numbers of the form $8(10 + 27n)$ are mischievous. Now the gap between two consecutive numbers of this form is $27(8) = 216$, so I could use this to show that $1/216 \approx 0.46\%$ of numbers are mischievous. We can use similar methods to show that all numbers of the form $27(37 + 125n)$ are mischievous, but this method seem to be giving us diminishing returns as it only gives that $1/3375 \approx 0.03\%$ of numbers are mischievous. Similarly all numbers of the form $125(3 + 8n)$ are mischievous which gives $1/1000 = 0.1\%$ of numbers being mischievous. Indeed if I just spam this trick we will eventually get to above 1% but it's definitely not the most practical solution.

It's natural to think about how far I can push this trick, and whether or not I can do this trick generally. Actually since our initial discovery gave us roughly halfway progress, I wondered if I could "flip" the signs of that equation to get another +0.5% progress. More precisely, the linear diophantine equation $27x + 8y$ also admits the solution

$$27(-5) + 8(17) = 1.$$

From this point we can use the same tricks as before to get that numbers of the form $27(5 + 8n)$ are mischievous. I was right this class of numbers gives another $1/216 \approx 0.46\%$ to the density of mischievous numbers. Finally to get that final $\approx 0.08\%$ I can just use the numbers $125(3 + 8n)$. This should be the problem solved, I'll write everything up cleanly tomorrow.

Solution

We will show that there are certain arithmetic progressions such that every term in those progressions is mischievous. Then we will show that these numbers appear often enough to be able to guarantee that at least $\approx 1.02\%$ of natural numbers are mischievous.

We begin with a couple of claims.

Claim 1: Every number of the form $8(-17 + 27n)$ is mischievous for $n \in \mathbb{N}$.

Proof: Let $k_n = 8(-17 + 27n)$, since $8 \mid k_n$, k_n is divisible by a perfect cube. $k_n + 1 = 216n - 135 = 27(8n - 5)$ and so $27 \mid (k_n + 1)$ and $(k_n + 1)$ is divisible by a perfect cube too. Thus k_n is always mischievous. $\square$

Claim 2: Every number of the form $27(-3 + 8n)$ is mischievous for $n \in \mathbb{N}$.

Proof: Let $k_n = 27(-3 + 8n)$, since $27 \mid k_n$, k_n is divisible by a perfect cube. $k_n + 1 = 216n - 80 = 8(27n - 10)$ and so $8 \mid (k_n + 1)$ and $(k_n + 1)$ is divisible by a perfect cube too. Thus k_n is always mischievous. $\square$

Claim 3: Every number of the form $125(-5 + 8n)$ is mischievous for $n \in \mathbb{N}$.

Proof: Let $k_n = 125(-5 + 8n)$, since $125 \mid k_n$, k_n is divisible by a perfect cube. $k_n + 1 = 1000n - 624 = 8(125n - 78)$ and so $8 \mid (k_n + 1)$ and $(k_n + 1)$ is divisible by a perfect cube too. Hence k_n is always mischievous. $\square$

Define three sequences given by

$$a_n = 8(-17 + 27n), \quad b_n = 27(-3 + 8n), \quad c_n = 125(-5 + 8n)$$

Claim 4: For any set A of consecutive natural numbers of size $N \geq 27(8) = 216$, there must be at least one term from a_n and at least one term from b_n in the set A .

Proof: For both sequences it is easy to verify that

$$a_{n+1} - a_n = 216, \quad b_{n+1} - b_n = 216.$$

I will prove the following auxiliary fact. Let $x \in \mathbb{N}$. If $x \equiv 80 \pmod{216}$ then x is a part of (a_n) . Note that $x \equiv 80 - 216 \equiv -136 \pmod{216}$. So we may write $x = 216n - 136 = 8(-17 + 27n)$, where $n \in \mathbb{N}$ because if $n < 1$ then this would imply that $x < 0$ which is impossible. Thus $x = a_n$ is a part of the sequence (a_n) .

Now among a set of $k \geq 216$ consecutive numbers there must be all residues modulo 216. Thus there is a number $\equiv 80$ modulo 216 and this number is a part of (a_n) .

We again prove an auxiliary result. Let $x \in \mathbb{N}$. If $x \equiv 135 \pmod{216}$, then x is a part of (b_n) . Observe that $x \equiv 135 - 216 \equiv -81 \pmod{216}$. Thus we can write $x = 216n - 81 = 27(-3 + 8n)$ where $n \in \mathbb{N}$ because if $n < 1$ then this would imply that $x < 0$ which isn't possible. Hence $x = b_n$ and is a part of the sequence (b_n) .

Now again among a set of $k \geq 216$ consecutive numbers there must be all residues modulo 216. Thus there must be a number $\equiv 135$ modulo 216 and this number is a part of (b_n) . $\square$

Claim 5: For any set A of consecutive natural numbers of size $N \geq 125(8) = 1000$, there must be at least one term from c_n in the set A .

Proof: Indeed one may check that

$$c_{n+1} - c_n = 1000.$$

Let's show the following auxilliary result. Let $x \in \mathbb{N}$. If $x \equiv 375 \pmod{1000}$, then x is a part of (c_n) . Observe that $x \equiv 375 - 1000 \equiv -625 \pmod{1000}$. So we can write $x = 1000n - 625 = 125(-5 + 8n)$ for some $n \in \mathbb{N}$ because otherwise if $n < 1$ then $x < 0$ which is impossible. Thus $x = c_n$ and is part of the sequence (c_n) .

Now among a set of $k \geq 1000$ consecutive numbers there must be all residues modulo 1000. Thus there is a number that is $\equiv 375 \pmod{1000}$, and this number is a part of (c_n) . $\square$

Claim 6: The following is true,

$$\{a_1, a_2, \dots\} \cap \{b_1, b_2, \dots\} = \emptyset$$

$$\{a_1, a_2, \dots\} \cap \{c_1, c_2, \dots\} = \emptyset$$

Proof: A number cannot be part of both (a_n) and (b_n) , since it would imply that the number is congruent to both 135, 80 modulo 216 which is impossible. Suppose some number k is a part of both (a_n) and (c_n) . Then $k = 1000m - 625$ for some $m \in \mathbb{N}$ and we require

$$1000m - 625 \equiv 80 \pmod{216}$$

$$136n \equiv 57 \pmod{216}.$$

Now if there is a solution to this congruence then there must exist $x, y \in \mathbb{Z}$ such that $136x + 216y = 57$. Bézout's theorem tells us that a solution will only exist if $\gcd(136, 216) = 8$ divides 57 which isn't true. $\square$

Claim 7: Let $k = 1000m - 625 = c_m$, ($m \in \mathbb{N}$) be a part of (c_n) . Then k is a part of (b_n) if and only if

$$m \in \{10, 37, 64, 91, 118, 145, 172, 199\} \pmod{216}.$$

Proof: For k to be a part of (b_n) we need,

$$1000m - 625 \equiv 135 \pmod{216}$$

$$136m \equiv 112 \pmod{216}$$

It suffices to study the linear diophantine equation

$$136m + 216y = 112$$

$$17m + 27y = 14$$

Now using whatever method you like, say Euclidean algorithm backtracking you can find that $17^{-1} \equiv 20 \pmod{27}$ and thus $m \equiv 14 \cdot 20 \equiv 10 \pmod{27}$. Now in mod 216 the only such m are

$$m \in \{10, 37, 64, 91, 118, 145, 172, 199\}. \square$$

Claim 8: In any group of 27 consecutive terms from (c_n) there is exactly one term from (b_n) .

Proof: Terms from (c_n) take on the form $1000n - 625$. With 27 consecutive terms, n takes on all residues modulo 27 exactly once. Now due to claim 7, a number is a part of (b_n) if $n \equiv 10 \pmod{27}$. Among these 27 consecutive terms exactly one has $n \equiv 10 \pmod{27}$ so exactly one term is a part of (b_n) . $\square$

Now let's take any set of $125(8)(27) = 27000$ consecutive natural numbers.

- In this set there are $\frac{27000}{216} = 125$ terms from (a_n) and $\frac{27000}{216} = 125$ terms from (b_n) , due to claim 4.
- In this set there are $\frac{27000}{1000} = 27$ terms from (c_n) , due to claim 5.

The numbers from these three sequences are all mischievous by claims 1,2,3. We know that the only times that terms can be a part of multiple sequences is when a term is a part of both (b_n) and (c_n) due to claim 6. Due to claim 7, we know that among those 27 terms from (c_n) , there is exactly one that is also a part of (b_n) , so we must take this into account.

Thus in any set of 27000 consecutive natural numbers there are at least $125 + 125 + 27 - 1 = 276$ mischievous numbers. Hence among these 27000 consecutive natural numbers, at least

$$\frac{276}{27000} = 1.02\% > 1\%$$

of the numbers are mischievous. ■

Comments: Really enjoyed working through this one!

Addendums (after having read solutions): Problem 7 was a really nice one. I guess this is an analysis problem? I essentially found the same solution as the official here, actually I think our constructions are almost exactly the same, the only thing is I didn't include numbers $\equiv 624 \pmod{1000}$, I think that's why my final density ended up dangerously close to 1%.

Problem 8

(a) Historically, it used to be possible to buy chicken McNuggets in boxes of 9 or 20. What was the largest number of chicken McNuggets that it was impossible to buy.

(b) If instead they were in boxes of size r and size s , with r and s coprime, show that the largest number that it is impossible to buy is $rs - r - s$. *This is actually called the Chicken McNugget Theorem!*

You might want to consider which numbers are possible for each value modulo s , if $s > r$.

(c) (Extension) Show that there are exactly $\frac{1}{2}(r-1)(s-1)$ positive integers that it is impossible to purchase.

Solution

(a) Say that we can express a natural number n *nicely* if there exists a solution to $n = 9x + 20y$, for $x, y \in \mathbb{Z}_{\geq 0}$. Let x, y be non-negative integers. It is possible to buy $9x + 20y$ chicken McNuggets. Bézout's theorem tells us that for $x, y \in \mathbb{Z}$ there is always a solution to $n = 9x + 20y$ since $\gcd(9, 20) = 1$. So for it to be "impossible" to buy some amount of chicken McNuggets, there must be some condition which forces one of $x, y < 0$.

Claim 1: Let $n = 9x + 20y$ be positive with $x, y \in \mathbb{Z}$. If $y \geq 9$ then we can always express n nicely.

Proof: If $x \geq 0$ then there is nothing to prove so assume that $x < 0$. For convenience let us write $a = -x > 0$. We can induct on y .

For the base case, note that if $n = 20(9) - 9a$, then $a < 20$ since $n > 0$. Now $n = -9(a - 20) = 9(20 - a)$, and since $20 - a > 0$ the base case is true.

Now assume the result is true for $y = k \geq 9$. That means that if $n = 20k - 9a$, then we can write it as $n = 9x' + 20y'$ for some $x', y' \in \mathbb{Z}_{\geq 0}$. Now consider,

$$\begin{aligned} m &= 20(k+1) - 9a \\ &= (20k - 9a) + 20 \\ &= (9x' + 20y') + 20 \\ &= 9x' + 20(y' + 1) \end{aligned}$$

and so the result is also true for $k+1$. Thus the proof is complete via induction. $\square$

Suppose that $N = 9x + 20y$ is the largest number of nuggets that it's impossible to buy. Due to claim 1, we know that $y \leq 8$. Since we want to find the largest such N we should just pick $y = 8$. Now we have $N = 20(8) + 9x$. Clearly if $x \geq 0$ then it is possible to buy N nuggets so we must have $x < 0$. Again since we want to maximise N it is best to pick $x = -1$, thus $N = 20(8) - 9(1) = 151$.

To finish we need to show that it isn't possible to express N nicely. Suppose $151 = 9x + 20y$ is written nicely. Taking mod 9 this becomes $2y \equiv 7 \pmod{9}$ from which you can use, say Bézout's theorem to show that $y \equiv 8 \pmod{9}$. If $y \geq 9$ then this forces $x < 0$, so this isn't possible. Also $y \geq 0$ by assumption so the only candidate for y is $y = 8$, from which we can solve to get $x = -1$, a contradiction. $\square$

(b) The general case is similar. We have $s, r \in \mathbb{N}$ with $\gcd(r, s) = 1$. Let us assume $s > r$.

Claim 1: Let $n = sX + rY$ be positive with $X, Y \in \mathbb{Z}$. If $X \geq r$ then we can express n nicely.

Proof: If $Y \geq 0$ then there is nothing to prove. So let us assume $Y < 0$. For convenience we will write $A = -Y > 0$. We are going to induct on X .

For the base case, we see that if $n = s(r) - rA$, then $A < s$ since n is positive. Now $n = r(s - A)$ and since $(s - A) > 0$ the base case is indeed true.

Now assume the result holds for $X = k \geq r$. Thus if $n = sk - rA$ we can write it as $n = sX' + rY'$ for some $X', Y' \in \mathbb{Z}_{\geq 0}$. Consider,

$$\begin{aligned} m &= s(k+1) - rA \\ &= (sk - rA) + s \\ &= (sX' + rY') + s \\ &= s(X' + 1) + rY' \end{aligned}$$

and so the result is true for $X = k + 1$ also. Thus the proof is complete via induction. $\square$

Suppose that $N = sX + rY$ is the largest number of nuggets that it's impossible to buy. Now due to claim 1, we must have $X < r$ and since we want to maximise N we will just pick $X = r - 1$. We have $N = s(r - 1) + rY$. Obviously if $Y \geq 0$ then N has been written nicely so $Y < 0$. Since we want to maximise N we will pick $Y = -1$. So we have $N = s(r - 1) - r = rs - s - r$.

To finish we need to show that it isn't possible to express N nicely. Suppose FTSOC that

$$N = rs - s - r = sX + rY$$

has been written nicely. Taking mod r we have

$$\begin{aligned} -s &\equiv sX \pmod{r} \\ -1 &\equiv X \pmod{r} \quad (*) \end{aligned}$$

(* — s has a modular inverse mod r because $\gcd(r, s) = 1$). If $X \geq r$ then the RHS will become too large and this will force $Y < 0$, so $X < r$. Also $X \geq 0$ since N was written nicely. Since $X \equiv -1 \pmod{r}$, these two inequalities mean that the only candidate for X is $X = r - 1$. But now plugging this in and solving yields $Y = -1$, a contradiction. $\blacksquare$

Remark: The solution I found is much nicer to follow if you consider mod r instead of mod s .

I'll write a little more detail about (). Let x be the modular inverse of a mod b . Then $ax \equiv 1 \pmod{b}$. This means that we can find $x, y \in \mathbb{Z}$ such that $ax + by = 1$. From Bézout's theorem we know that solutions only exist if $\gcd(a, b) \mid 1$, that is $\gcd(a, b) = 1$.*

(c) We'll take the same setup as in part (b). That is let $s, r \in \mathbb{N}$ with $\gcd(r, s) = 1$ and $s > r$. Similar to claim 1, in part (b) it is natural to guess that the following result is also true.

Claim 1: Let $n = sX + rY$ be positive with $X, Y \in \mathbb{Z}$. If $Y \geq s$ then we can express n nicely.

Proof: If $X \geq 0$ there is nothing to prove, so assume $X < 0$. For convenience let $A = -X > 0$. We will induct on Y .

For the base case when $Y = s$, we have $n = r(s) - sA$ with $A < r$ since $n > 0$. Now $n = s(r - A)$ and since $r - A > 0$ the base case is true.

Now assume that the result holds for $Y = k \geq s$. So if $n = rk - sA$ we can write it as $n = sX' + rY'$ for some $X', Y' \in \mathbb{Z}_{\geq 0}$. Consider

$$\begin{aligned} m &= r(k + 1) - sA \\ &= (rk - sA) + r \\ &= (sX' + rY') + r \\ &= sX' + r(Y' + 1) \end{aligned}$$

and so the result is also true for $k + 1$. The proof is finished via induction. $\square$

Let $n = sX + rY$ be positive and impossible to write nicely. Exactly one of X, Y must be negative. This is because if they were both negative then this would mean that $n < 0$, and if they were both ≥ 0 then n has been written nicely.

Claim 2: For the n described above, we can assume $Y < 0$.

Proof: We will show that any such n with $X < 0$ can be re-written in such a way that $Y < 0$. Suppose indeed that $Y > 0, X < 0$, let $A = -X > 0$ for convenience. So we have $n = rY - sA$ being positive and impossible to write nicely. Due to claim 1 (c) we must have $Y < s$. Also since

$$0 < n = rY - sA < rs - sA$$

we have $sA < rs \Rightarrow A < r$.

$$\begin{aligned} n &= rY - sA \\ &= rY - sA + (rs - rs) \\ &= r(Y - s) - s(A - r) \\ &= s(r - A) + r(Y - s) \end{aligned}$$

but thanks to the work above we know that $Y - s < 0$ and $r - A > 0$, so if we just let $X' = r - A$ and $Y' = Y - s$, we have written $n = sX' + rY'$ with $X' > 0, Y' < 0$. $\square$

Now thanks to claim 2, we safely assume $X > 0$ and $Y < 0$. Due to claim 1 (b) we have

$$X \in \{1, 2, \dots, r - 1\}$$

Let $B = -Y > 0$ for convenience. We want to find the set of values that B can take on. Since $n > 0$,

$$\begin{aligned} n &= sX - rB > 0 \\ sX &> rB \\ \frac{sX}{r} &> B \\ B &\leq \left\lfloor \frac{sX}{r} \right\rfloor \end{aligned}$$

where the last line is because $B \in \mathbb{Z}$. So for each selection of X there are $\lfloor sX/r \rfloor$ choices for B or equivalently

$$B \in \left\{ 1, 2, \dots, \left\lfloor \frac{sX}{r} \right\rfloor \right\}.$$

Thus putting this together there are precisely

$$S = \sum_{X=1}^{r-1} \left\lfloor \frac{sX}{r} \right\rfloor$$

integers that are impossible to write nicely. So now our goal is to evaluate S . Though it looks like a nasty sum, it is possible to evaluate it nicely. We need a key lemma.

Claim 3: For $X = 1, 2, \dots, r-1$, the number sX leaves all non-zero residues modulo r . Precisely that is

$$\{s(1), s(2), \dots, s(r-1)\} = \{1, 2, \dots, r-1\}.$$

where all those numbers are reduced modulo r .

Proof: Suppose FTSOC that there is some integer $1 \leq q \leq r-1$, such that there does not exist $X = 1, 2, \dots, r-1$ with

$$sX \equiv q \pmod{r}$$

Since $\gcd(s, r) = 1$ we can write

$$X \equiv s^{-1}q \pmod{r}$$

Now the only way that X cannot exist in this scenario is if $s^{-1}q \equiv 0$. However $q > 0$ and the modular inverse of s is obviously not 0, so we have reached a contradiction. $\square$

Now we can write

$$\begin{aligned} S &= \sum_{X=1}^{r-1} \frac{sX}{r} - \left\{ \frac{sX}{r} \right\} \\ &= \frac{s(r-1)r}{2} - \sum_{X=1}^{r-1} \left\{ \frac{sX}{r} \right\} \\ &= \frac{s(r-1)}{2} - \sum_{X=1}^{r-1} \left\{ \frac{sX}{r} \right\} \end{aligned}$$

Now thanks to claim 3, sX leaves every non-zero residue mod r . Since the fractional part of $\frac{sX}{r}$ is just $\frac{sX \bmod r}{r}$ we can write

$$\begin{aligned}
 &= \frac{s(r-1)}{2} - \sum_{k=1}^{r-1} \frac{k}{r} \\
 &= \frac{s(r-1)}{2} - \frac{1}{r} \frac{(r-1)r}{2} \\
 &= \frac{s(r-1)}{2} - \frac{(r-1)}{2} \\
 &= \frac{1}{2}(s-1)(r-1). \blacksquare
 \end{aligned}$$

Comments: This one was really nice also, lots of moving parts here. Throughout that solution $\lfloor \cdot \rfloor$ refers to the floor of a number, and $\{ \cdot \}$ refers to the fractional part of a number, (or the brackets enclose a set, it is dependent on context.)

Addendums (after having read solutions): My solutions (once again) look very different to the official ones. It seems that the official solutions are much more succinct and avoid using much technical language, which is understandable since I guess UKMT wants the solutions to be understandable even to those who have only the tools at GCSE maths level. I think a nice way to describe it is that the UKMT solutions are more "philosophical" as in they reason with pure logic why the answer should be something, whilst my solutions are more explicit or computational showing why an answer should be true with concrete calculations and theorems.

Though my solution is much longer, I particularly liked my solution to part (c) here. At first it seems really impossible to evaluate S as it doesn't look like any of the standard summations that we have learnt to deal with, but a nice solution does in fact exist using the floor and fractional part functions.